4 Bloom Filter: Reduce the memory needed to keep track of content

This chapter covers

- Describing and analyzing *Bloom filters*
- Solving a problem: keeping track of large documents using little memory
- Showing why dictionaries are an imperfect solution
- Improving the memory print by using *Bloom filters*
- Recognizing use cases where Bloom filters improve performance
- Using metrics to tune the quality of *Bloom filters'* solution

Starting with this chapter we'll be reviewing less common data structures that solve - as strange as it might seem - common problems. *Bloom filters* are one of the most prominent examples: they are widely used in most